

Recall

1. Every nls has a completion.
2. Examples of nls with completion.
3. Quotient spaces.

• $X(\mathbb{K}) \rightarrow M$ subspace

$$x, y \in X$$

$$x \sim y \Leftrightarrow x - y \in M$$

• ' $\sim$ ' is a equivalence relation.

$$[x] = x + M$$

$$\begin{aligned} X/M &= \{ [x] \mid x \in X \} \\ &= \{ x + M \mid x \in X \} \end{aligned}$$

'+' , '·' well defined.

$$(x+M) + (y+M) = (x+y)+M$$

$$\|x+M\| = \text{dis}(x, M)$$

$$(\text{dist}(x, M) = 0 \Leftrightarrow x \in \overline{M})$$

$$M \subseteq \overline{X}$$

closed

$$\|\cdot\| \text{ on } X/M$$

Theo Let X be a Banach space
and M be a closed subspace
of X . Then X/M is a
Banach space.

Lemma

Let (y_n) is a C.S in
a NLSY.

Then $\exists$ a subseq (y_{n_k}) of
 (y_n) s.t

$$\|y_{n_{k+1}} - y_{n_k}\| < 2^{-k}.$$

Proof

Given that (y_n) is
a C.S

By defn.

$\forall \epsilon > 0 \exists N$ s.t

$$\|y_m - y_n\| < \epsilon \quad \forall n, m \geq N.$$

Choose $\epsilon = \frac{1}{2}$ then $\exists n_1$ s.t

$$\|y_m - y_n\| < \frac{1}{2} \quad \forall m, n \geq n_1,$$

then for $\epsilon = \frac{1}{2^2}$ $\exists n_2$ s.t

$$n_2 > n_1,$$

$$\& \quad \|y_m - y_n\| < \frac{1}{2^2} \quad \forall m, n \geq n_2$$

Continuing we get

$$\|y_m - y_n\| < \frac{1}{2^k} \quad \forall m, n \geq n_k$$

$$\& \quad n_1 < n_2 < n_3 \dots < n_k:$$

Put $m = n_{k+1}$ $n = n_k$

we get the claim.

Claim : X/M is a Banach space.

Let $(x_n + M)$ be a Cauchy seq in X/M .

Claim': $\exists (x_{n_k})$ s.t. $(x_{n_k} + M)$ is convergent.

Since

X/M is a nls so for $(x_n + M) \exists$ a subseq $(x_{n_k} + M)$ s.t.

$$\| (x_{n_{k+1}} + M) - (x_{n_k} + M) \| < \frac{1}{2}\epsilon.$$

①.

Consider $y_1 = 0$

$$\inf_{y \in M} \| (x_{n_1} - y_1) - (x_{n_2} - y) \|$$

$$\geq \inf_{y \in M} \| (x_{n_1} - x_{n_2}) + y \|$$

$$\geq \| x_{n_1} - x_{n_2} + M \|$$

$$< \frac{1}{2}$$

$\exists y_2$ s.t.

$$\| (x_{n_1} - y_1) - \underbrace{(x_{n_2} - y_2)} \|$$

$$< 2 \cdot \frac{1}{2}$$

(Using defn of infimum)

Since $y_2 \in M$.

$$\inf_{y \in M} \| (x_{n_2} - y_2) - (x_{n_3} - y) \|$$

$$= \inf \| (x_{n_2} - x_{n_3}) + y \|$$

$$\left(\text{dist } (x, M) = \text{dist } (x+y, M) \right. \\ \left. y \in M \right)$$

$$= \| x_{n_2} - x_{n_3} + M \|$$

$$< \frac{1}{2}^2.$$

By defn. of infimum $\exists y_3 \in M$
s.t

$$\| (x_{n_2} - y_2) - (x_{n_3} - y_3) \|$$

$$< 2 \cdot \frac{1}{2^2}$$

Continuing like this $\exists y_k$ s.t

$$\| (x_{n_k} - y_k) - (x_{n_{k+1}} - y_{k+1}) \|$$

$$< 2 \cdot \frac{1}{2^k}.$$

$$z_k = x_{n_k} - y_{n_k}.$$

* We can say

$$\|z_k - z_{k+1}\| < \frac{1}{2^{k-1}}$$

Then (z_k) is a C. Seq. in X

$$\left[\begin{array}{l} m > n \\ \|z_m - z_n\| < \epsilon \quad \forall n \geq N \end{array} \right]$$

So $\underline{z_k} \rightarrow \underline{z}$ as $k \rightarrow \infty$
 $z \in X$.

$$\begin{aligned} & \| (x_{n_k} + M) - (z + M) \| \\ &= \| (x_{n_k} - z) + M \| \end{aligned}$$

$$= \| \underbrace{x_{n_k} - y_k}_{= z_k} - z + M \|$$

$$= \| z_k - z + M \|$$

$$\left(= \inf \| z_k - z + u \|, u \in M \right)$$

$$\leq \| z_k - z \| \rightarrow 0 \text{ as } k \rightarrow \infty$$

$$\therefore (x_{n_k} + M) \xrightarrow[k \rightarrow \infty]{} (z + M) \text{ as}$$

Converse

Theo. Let X be a nls
 $\&$ M a ~~also~~ closed subspace

of X .

If X/M is Banach & M is also Banach then X is Banach.

Proof: Let (x_n) be a C.S in X .

$$\text{Then } \|x_n + M - x_m + M\|$$

$$= \|x_n - x_m + M\|$$

$$= \inf \{ \|x_n - x_m + u\| \mid u \in M \}$$

$$\leq \|x_n - x_m\| \rightarrow 0 \text{ as } m, n \rightarrow \infty$$

$\therefore (x_n + M)$ is a C.S in X/M .

Since X/M is a Banach space

$$\Rightarrow (x_n + M) \rightarrow (x + M) \quad x \in X.$$

$\text{as } n \rightarrow \infty.$

$$\| (x_n + M) - (x + M) \| < \epsilon \quad \forall n \geq N.$$

$$\Rightarrow \underline{\| (x_n - x) + M \|} < \epsilon \quad \forall n \geq N.$$

$$\| (x_n - x) + M \| = \inf \{ \| x_n - x + y \| \mid y \in M \}$$

Choose $y_n \in M$ s.t

$$\| x_n - x + y_n \| \leq \| (x_n - x) + M \| + \frac{1}{n}$$

$\rightarrow 0 \quad \text{as } n \rightarrow \infty$

$$\therefore \quad (\underline{x_n + y_n}) \longrightarrow x \quad \text{in } X$$

~~Then~~ Claim (y_n) is a C.S. in M
(exercise)

Convergent series in NLS.

Let (x_n) be a seq in a NLS, X .

Consider $\sum_{n \in \mathbb{N}} x_n$.

This series converges to x

$$S_N = \sum_{n=1}^N x_n \longrightarrow x \quad \text{in } X.$$

$$\|S_N - x\|_x \rightarrow 0 \text{ as } N \rightarrow \infty.$$

Absolutely convergent series

If $\sum \|x_n\| < \infty$ then $\sum x_n$ is absolutely convergent.

Ex.
 $\mathbb{R}$.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} \rightarrow \text{convergent.} \quad \begin{matrix} \| \cdot \| \\ = 1 \end{matrix}$$

but $\sum_{n=1}^{\infty} |(-1)^{n+1} \frac{1}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ is not convergent.

Remark:

Notion of abs convergence is a NLS is NOT stronger than convergence.

Ex $X = \mathcal{P}([0,1])$, $\|\cdot\|_\infty$

$$\sum_{n=1}^{\infty} \frac{x^n}{n!}$$

Here $\sum_{n=1}^{\infty} \left\| \frac{x^n}{n!} \right\|$ is convergent

$\therefore \sum \frac{x^n}{n!}$ is ~~con~~ absolutely convergent

However $\sum \frac{x^n}{n!} \Rightarrow e^x \notin \mathcal{P}[0,1]$

$\therefore$ This is NOT convergent.

Theo.

Let X be NLS.

Then X is a Banach space

$\Leftrightarrow$ Every ~~con~~ absolutely

convergent series in X is convergent.

Proof: Suppose X is a Banach space.

Let $\sum x_n$ be absolutely convergent.

(ie $\sum \|x_n\|$ is convergent)

Let $\sum x_n$ be an absolutely convergent series.

ie. $\sum \|x_n\|$ is convergent.

Let $S_N = \sum_{n=1}^N x_n$. $N > M$

$$\|S_N - S_M\| = \left\| \sum_{n=1}^N x_n - \sum_{n=1}^M x_n \right\|$$

$$\leq \sum_{n=M+1}^N \|x_n\|$$

$$\|S_N - S_M\| \rightarrow 0 \quad \text{as} \quad N, M \rightarrow \infty.$$

$\therefore (S_N)$ is a C. Seq.

Since X is Banach so
 (S_N) is convergent and

that implies $\sum x_n$ is convergent.